



Deep Learning Optimized on Jean Zay

Dataset optimization

Storage spaces and data format



IDRIS



Dataset optimization

Main bottlenecks ◀

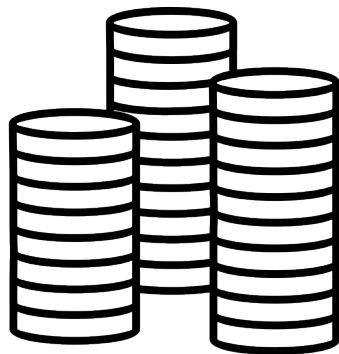
Data storage – various disk spaces ◀

Data format – at sample level ◀

Data format – at dataset level ◀

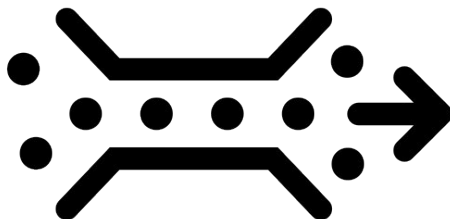
Bottlenecks upstream of DataLoader

Storage disks



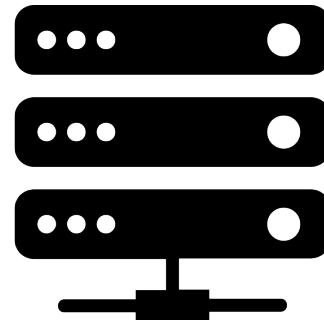
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

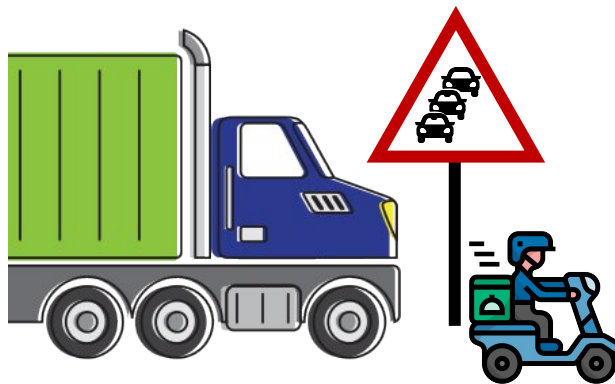
Bottlenecks upstream of DataLoader

Storage disks



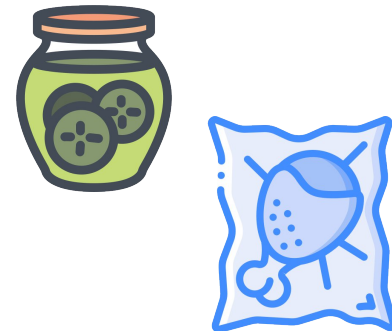
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

Dataset optimization

Main bottlenecks ◀

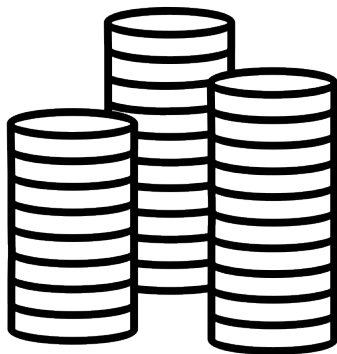
Data storage – various disk spaces ◀

Data format – at sample level ◀

Data format – at dataset level ◀

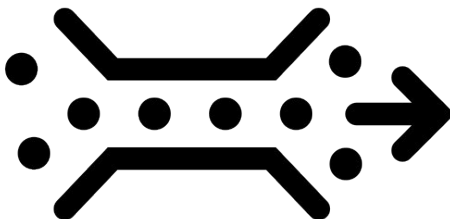
Bottlenecks upstream of DataLoader

Storage disks



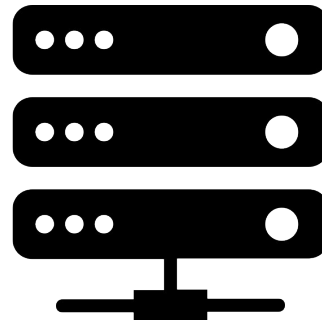
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

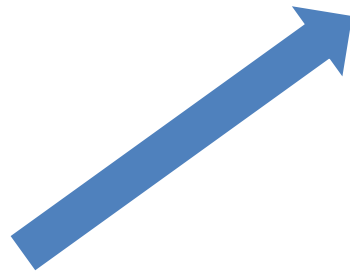
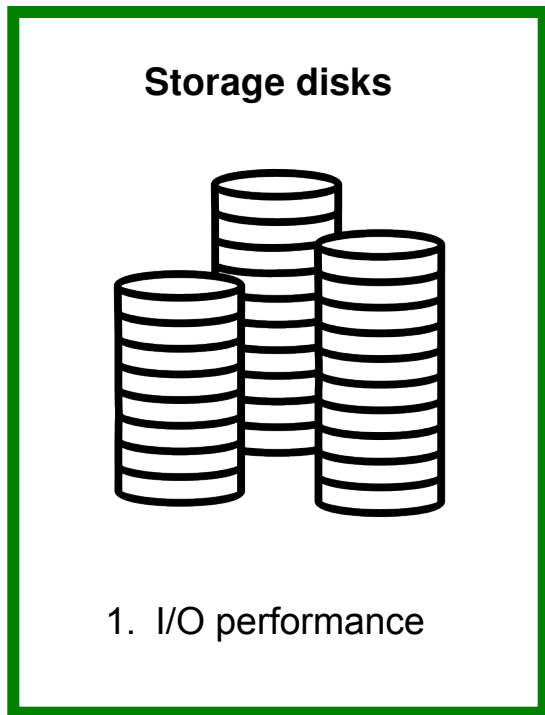
CPU workers



3. Decoder performance

Where should I store my dataset?

Various disk spaces



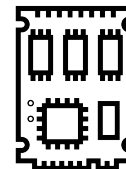
WORK
Rotative disk spaces



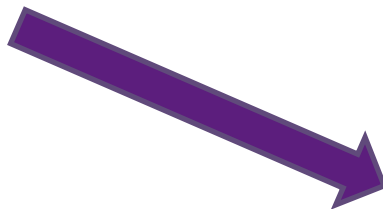
300-350 GB/s



SCRATCH
Full Flash



1.1-1.5 TB/s



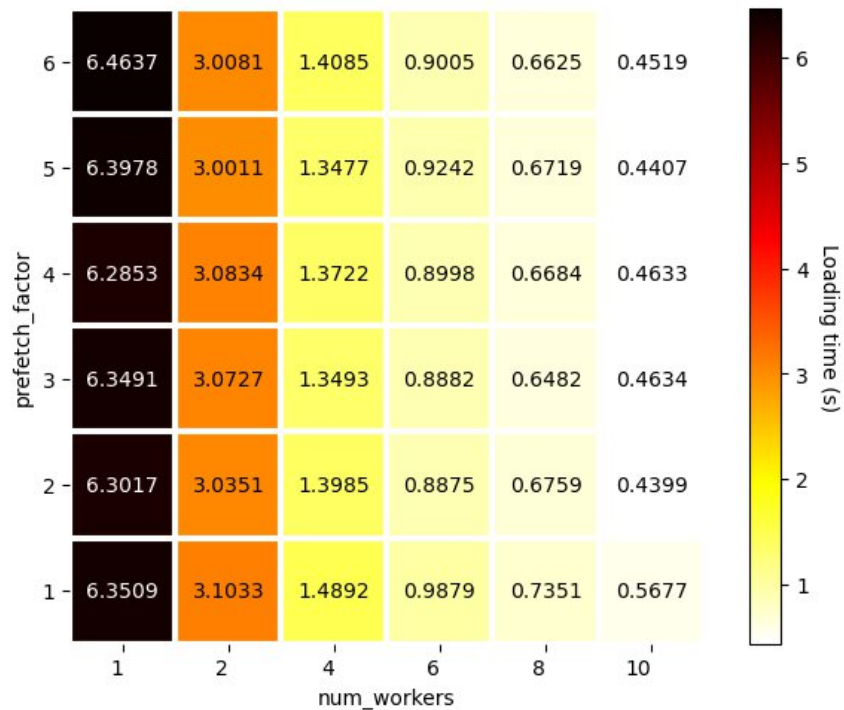
NVMe
Local disk
(test configuration)



Local

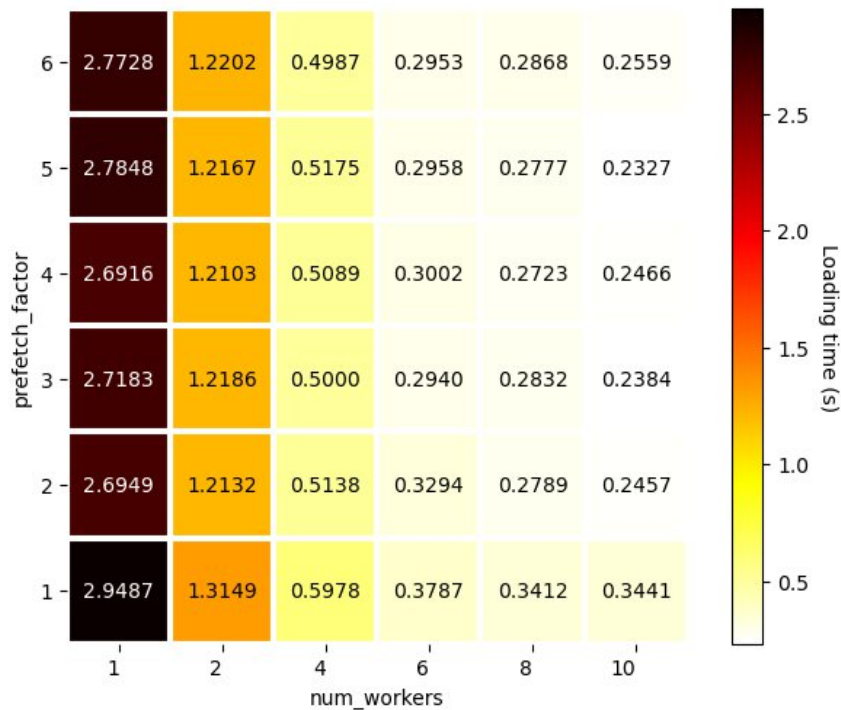
Various disk spaces

dlojz.py - 50 iterations - test partition gpu_p4



WORK

100 GB/s - OPA

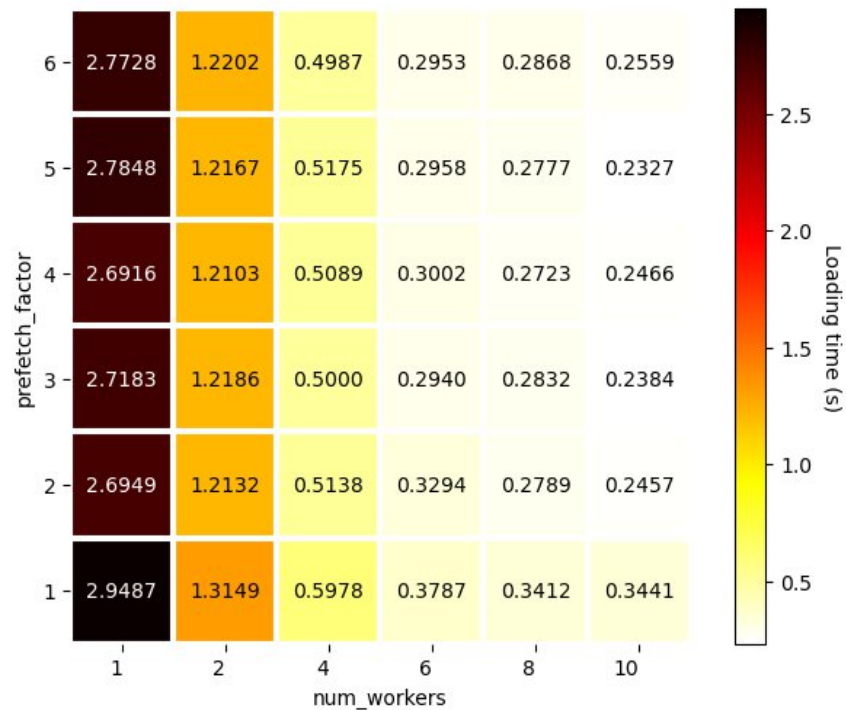


SCRATCH

500 GB/s - OPA

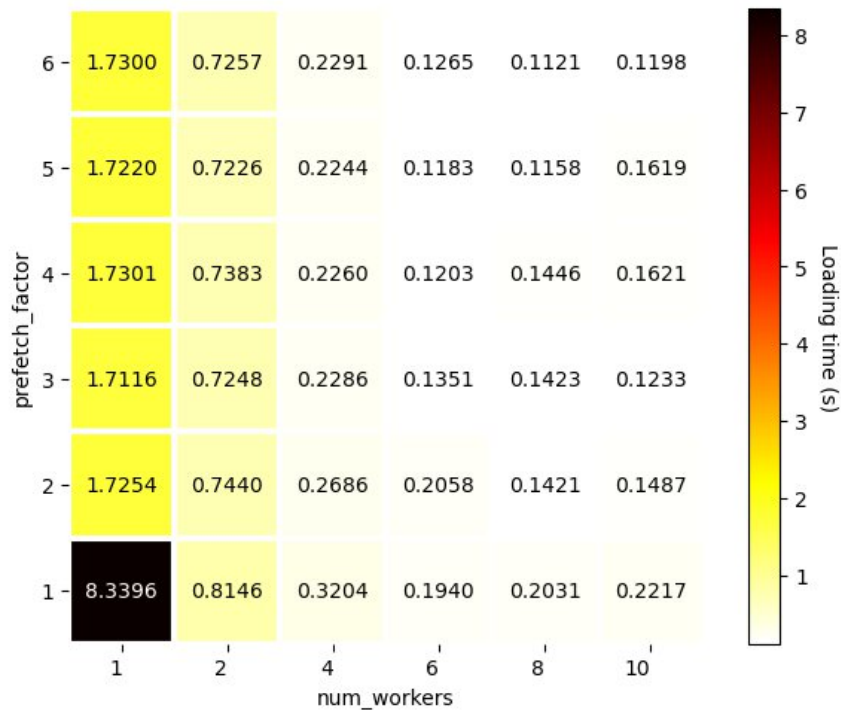
Various disk spaces

dlojz.py - 50 iterations - test partition gpu_p4



SCRATCH

500 GB/s - OPA



NVMe

PCIe

Various disk spaces

- **NVMe**
 - ✓ Best IO performance
 - You need to copy your dataset on the local disk first, which can take a very long time
 - This solution is not suitable at the scale of a supercomputer so it is not available to users
- **SCRATCH**
 - ✓ Second best IO performance
 - ✓ Very large quota (bytes and inodes)
 - 30 days file lifespan
- **WORK**
 - Worst performance (but not bad)
 - Only 5 TB and 500k inodes
 - ✓ No automatic deletion procedure

Various disk spaces

- **What about the STORE?**
 - Magnetic tape storage (with a cache on rotative disk)
 - 50 TB and 100k inodes
 - Intended to receive few very large-sized files (up to 10 TiB per file) → **archives**
 - Not accessible from the compute nodes (very high read/write latency)
- **Good practice**
 - If you don't need IO performance (when building your workflow, debugging, or simply if IO is not a bottleneck), store your dataset in the **WORK** if it is not too large
 - If you need IO performance or if it is too large, store your dataset in the **SCRATCH**
 - Keep a duplicate of your dataset **as an archive in the STORE**

Dataset optimization

Main bottlenecks ◀

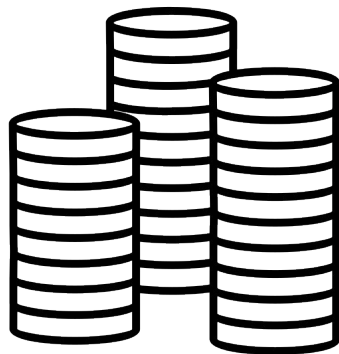
Data storage – various disk spaces ◀

Data format – at sample level ◀

Data format – at dataset level ◀

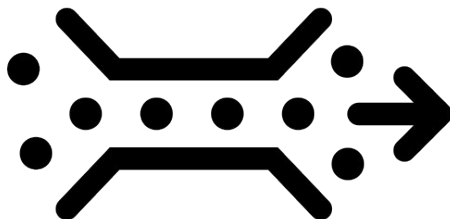
Bottlenecks upstream of DataLoader

Storage disks



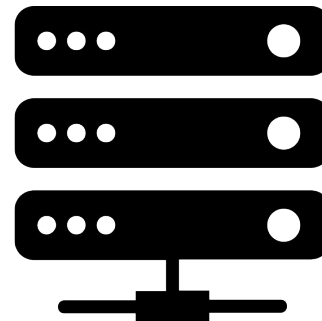
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

Which format for my data?

At sample level - Sample decoding



Binary format: Pickle format, HDF5,...
Decoded more quickly, takes more space

- ✓ Decoder performance
- Shared bandwidth
- Storage volume



Compressed format: jpeg, png,...
Decoded more slowly, takes less space

- Decoder performance
- ✓ Shared bandwidth
- ✓ Storage volume

Dataset optimisation

Main bottlenecks ◀

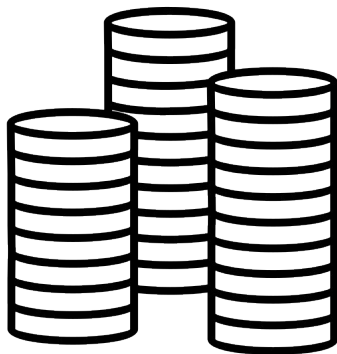
Data storage – various disk spaces ◀

Data format – at sample level ◀

Data format – at dataset level ◀

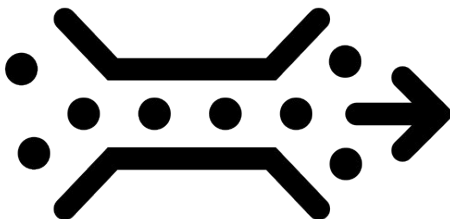
Bottlenecks upstream of DataLoader

Storage disks



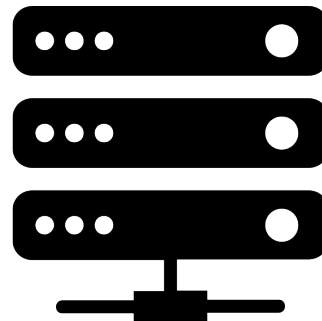
1. I/O performance

Interconnection network
Omnipath/InfiniBand



2. Shared bandwidth

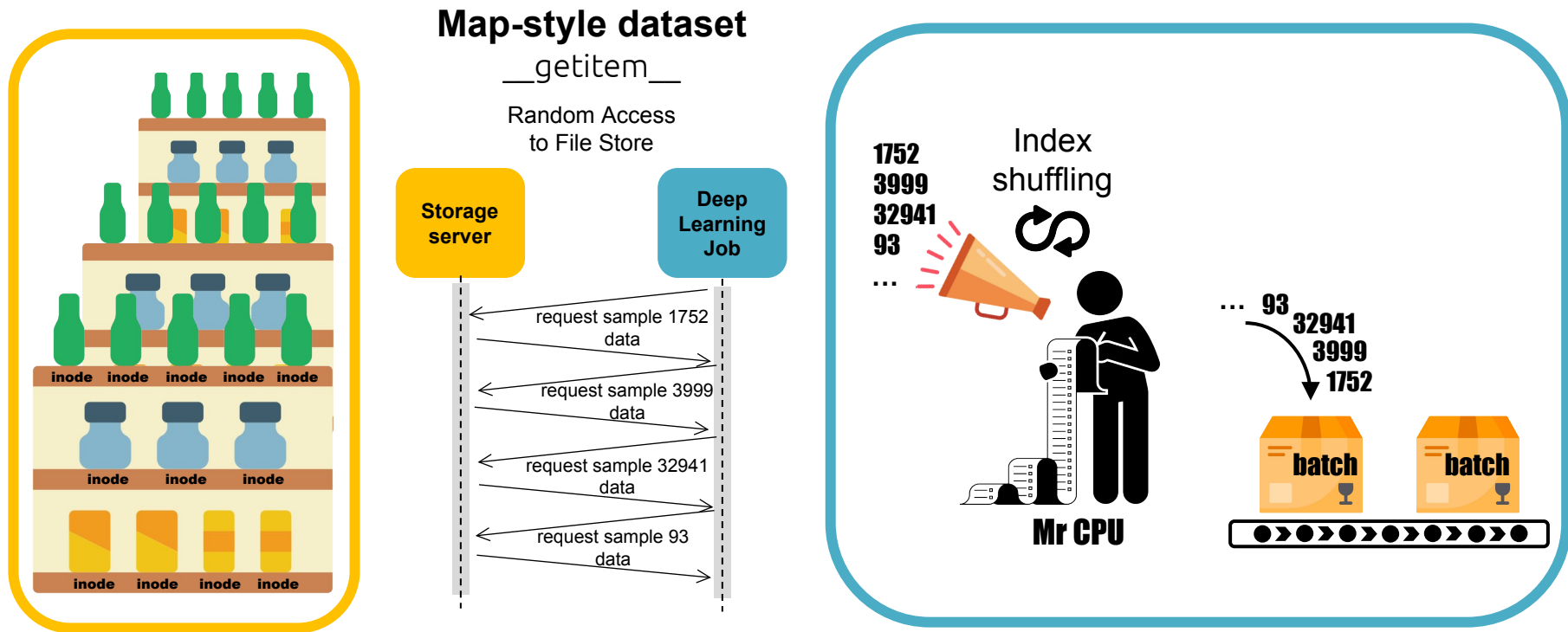
CPU workers



3. Decoder performance

Which format for my dataset?

Intuitive way: 1 sample = 1 file



Pros: Easy to handle, random access possible

Cons: Lots of inodes, lots of I/Os

Too many inodes is an issue

Error: Disk quota exceeded



- \$WORK quota per user is 5 TB / **500 kinodes**
- \$SCRATCH safety quota per user is ~500TB / **150 Minodes**

+ **Parallel file system does not like:**

- intensive IO workloads involving lots of small files
- large number of files in a single directory



Too many inodes is an issue

- IDRIS recommendation on Jean Zay: **less than 100k inodes per folder**

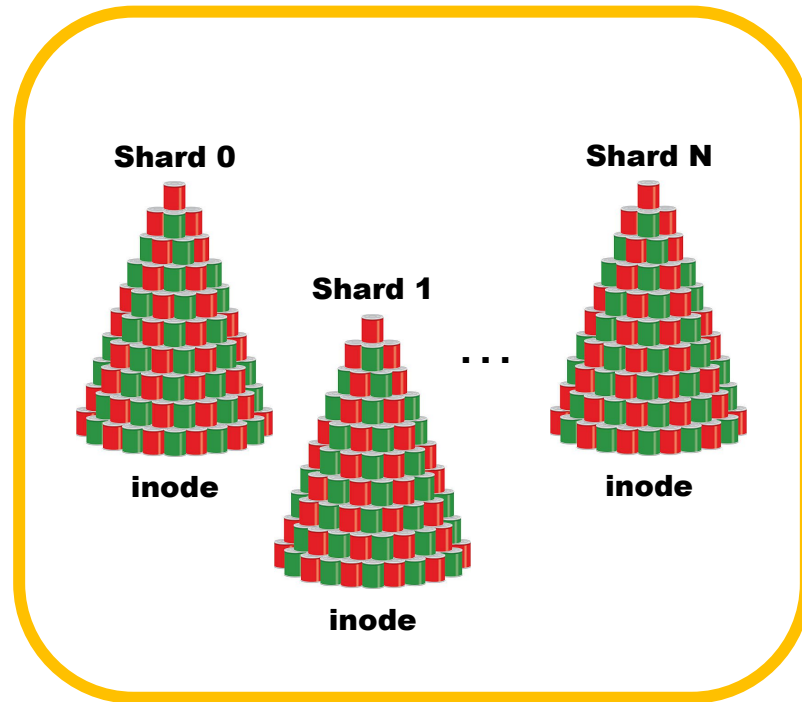
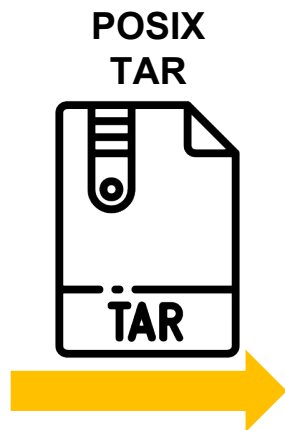
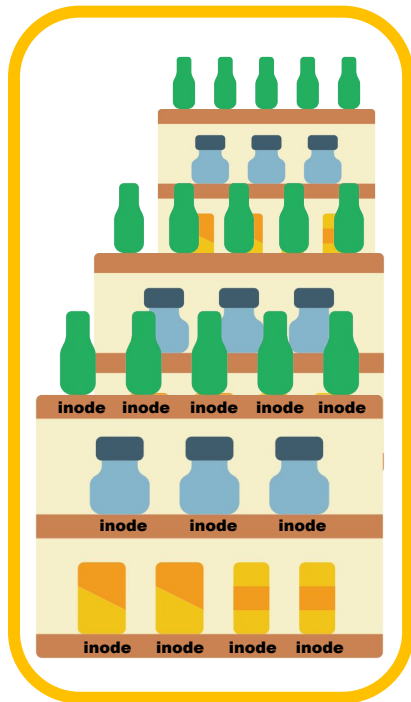
- Repository limitations and recommendations on HuggingFace Hub 🙌

Characteristics	Recommended	Tips
Files per repo	<100k	merge data into fewer files
Entries per folder	<10k	use subdirectories in repo

<https://huggingface.co/docs/hub/storage-limits#repository-limitations-and-recommendations>

+ for large datasets, **WebDataset** format is recommended

WebDataset format – Gathering data



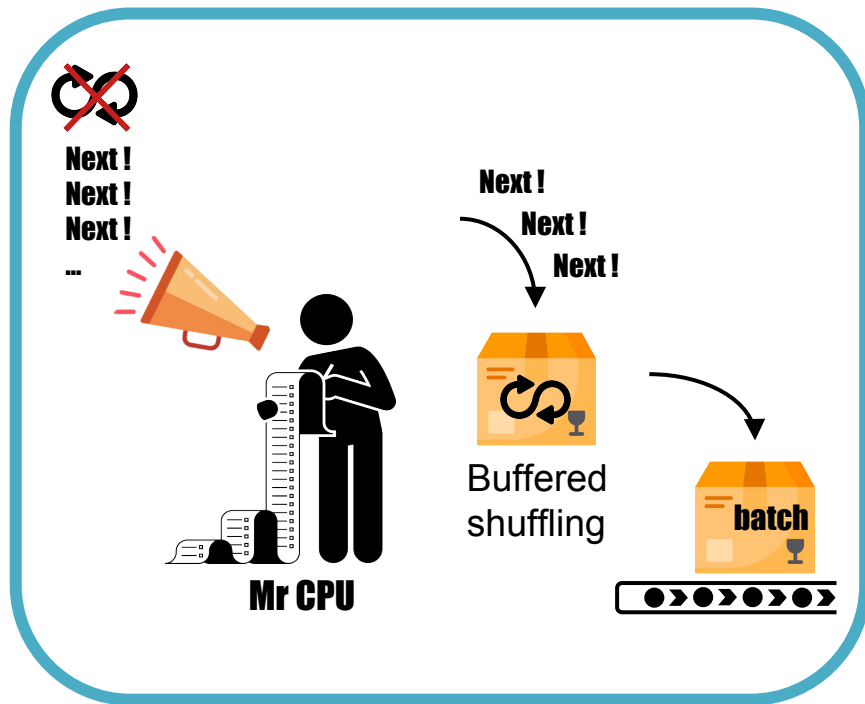
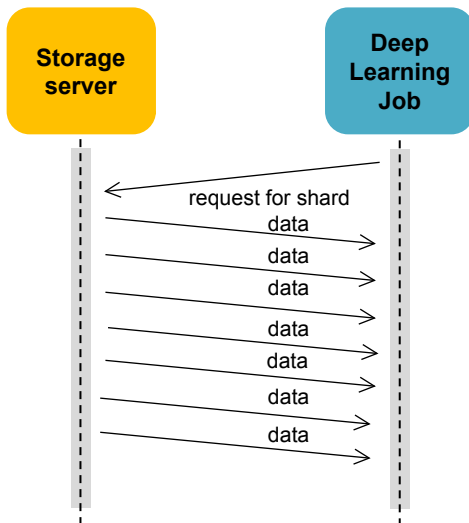
WebDataset format – Iterable dataset



Iterable-style dataset

`__iter__`

Pipelined Access
to Object Store

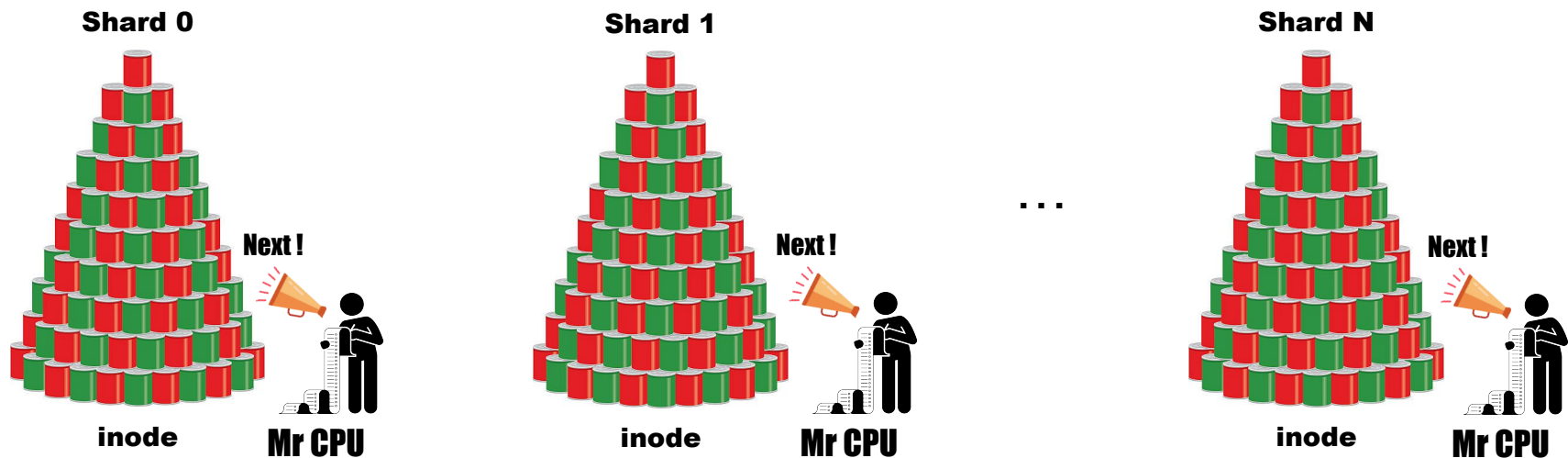


Pros: Fewer I/Os, fewer inodes

Cons: **Preprocessing**, difficult to shuffle or distribute, unknown dataset length

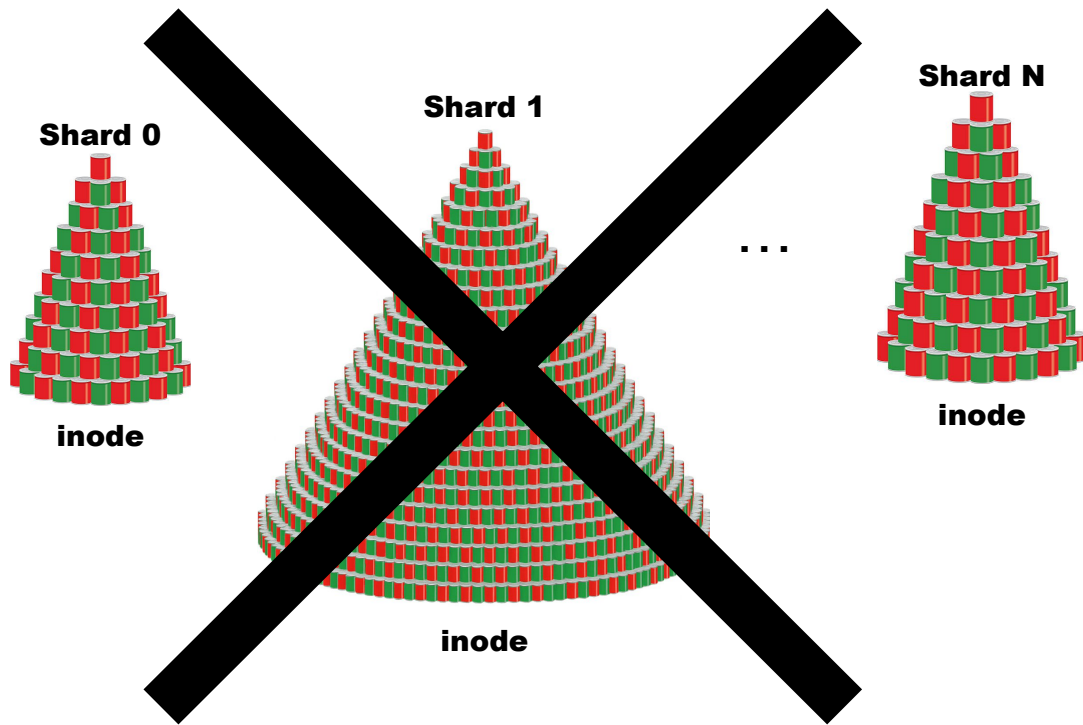
WebDataset format - Sharding

Sharding is necessary to benefit from parallel implementation
(DataLoader multi-processing and Distributed Data Parallelism).



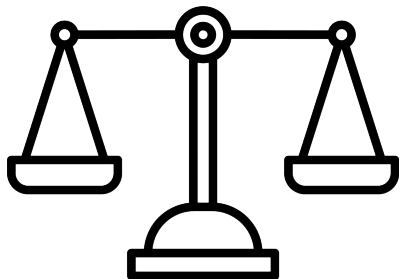
The number of shards should be a multiple of the number of tasks/GPUs.

WebDataset format - Sharding

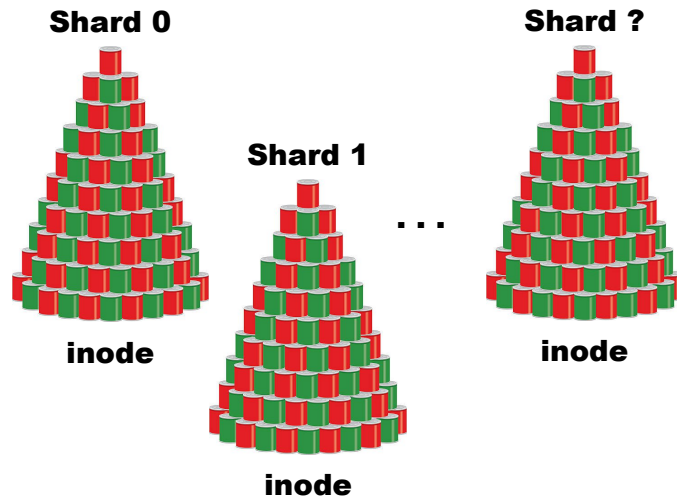


Samples must be evenly distributed among the shards to balance the workload between processes.

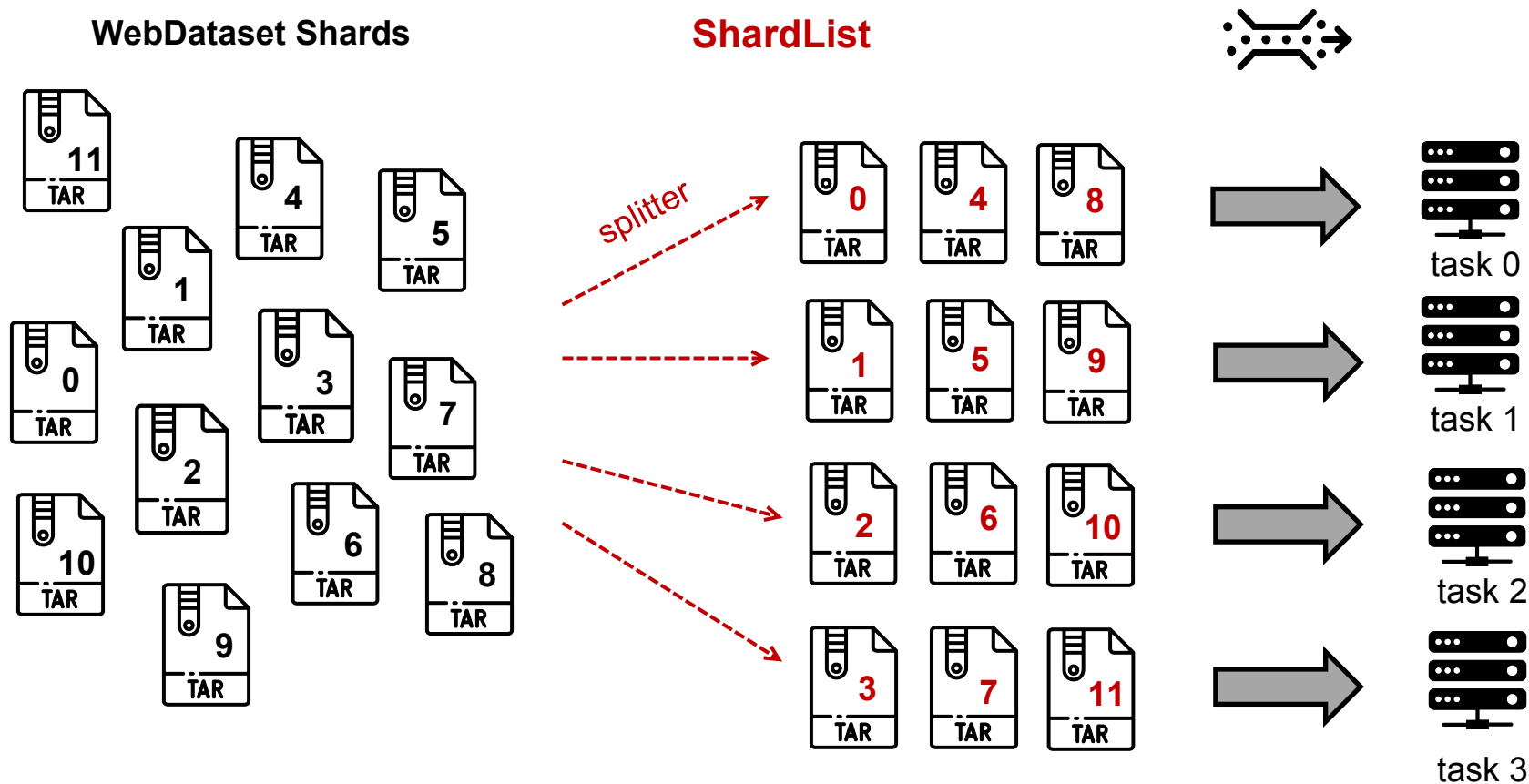
WebDataset format - More or less shards?



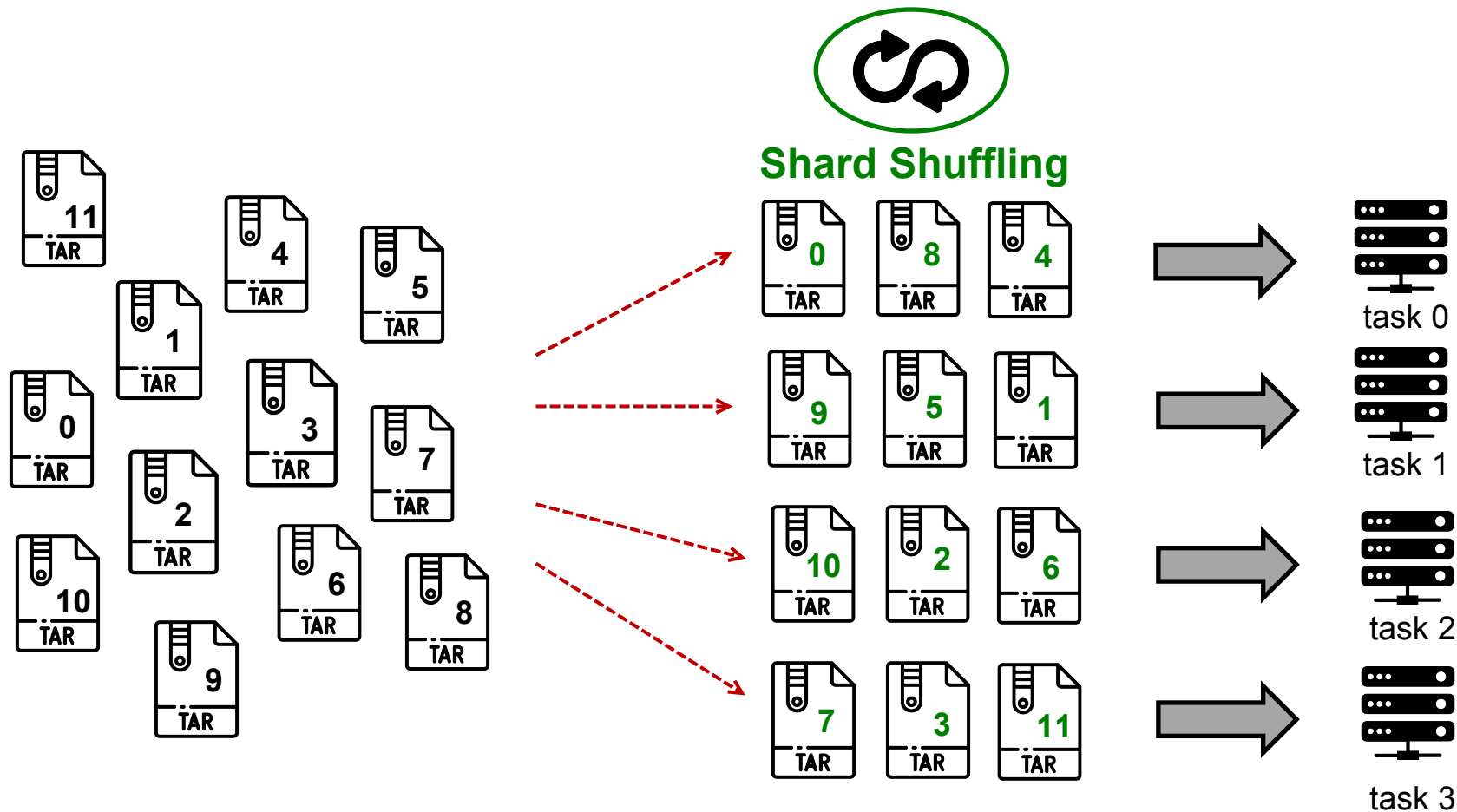
	More shards	Less shards
Large scale distribution	+	-
Shared bandwidth	+	-
Inodes quota	-	+
Number of I/O	-	+



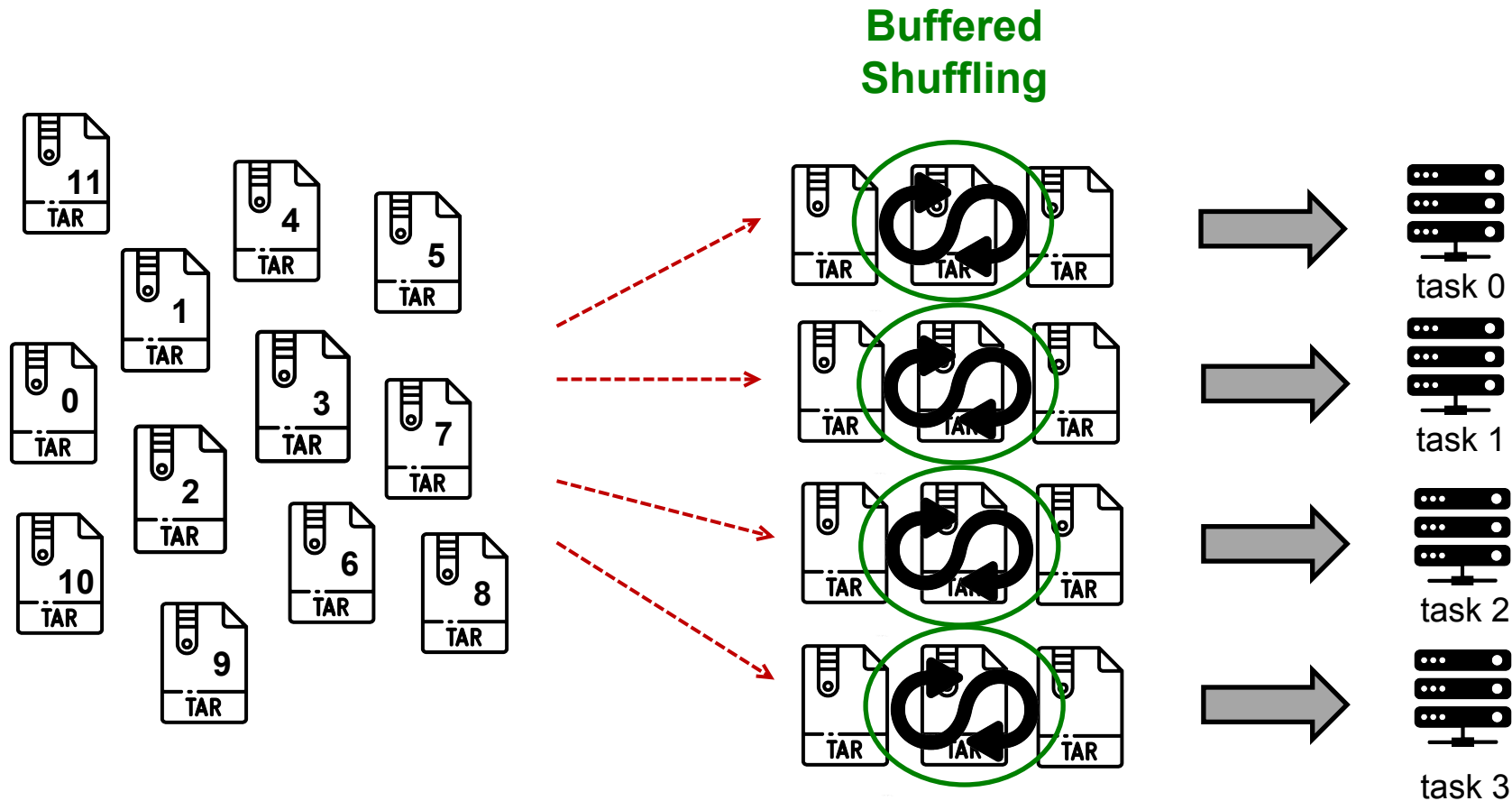
WebDataset – Multiworker sharding



WebDataset - Shuffling

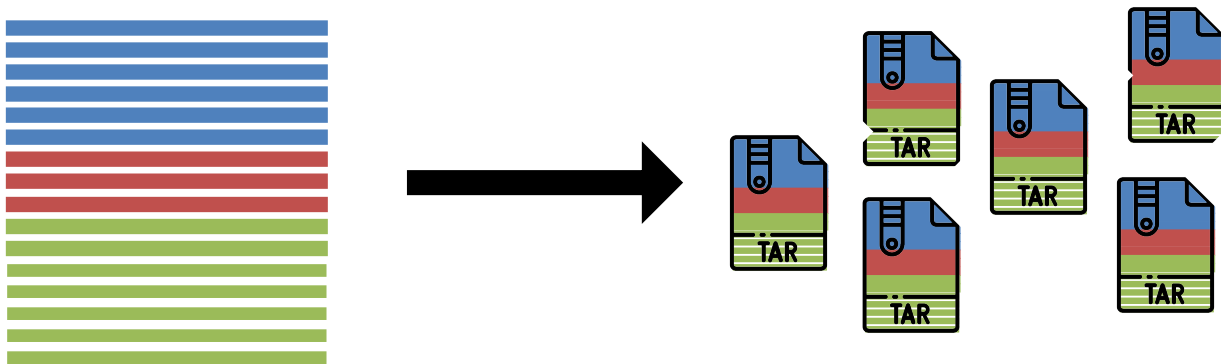


WebDataset - Shuffling



WebDataset - Preprocessing

- Distribute the samples as **evenly** as possible among the shards.
- Choose the number of shards **according to the number of GPUs & DataLoader workers** you will use.
- Distribute the samples so that each shard contains a **representative part of the dataset**.



+ Converting data before creating the archives to improve decoding performance?



WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds
```

```
def my_splitter(paths):  
    paths = list(paths)  
    return paths[idr_torch.rank::idr_torch.world_size]
```

} distribute shards among processes

```
paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'  
train_dataset_len = 1281167  
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True) \  
    .shuffle(1000) \  
    .decode("torchrgb") \  
    .to_tuple('input.pyd', 'output.pyd') \  
    .map_tuple(transform, lambda x: x) \  
    .batched(mini_batch_size) \  
    .with_length(train_dataset_len)
```

```
nbatches = train_dataset_len // global_batch_size  
train_loader = wds.WebLoader(train_dataset, batch_size=None, \  
    num_workers=num_workers, \  
    persistent_workers=persistent_workers, \  
    pin_memory=pin_memory, \  
    prefetch_factor=prefetch_factor \  
    ).slice(nbatches)
```

```
train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```



shuffling shards indexes
per process

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000) \
    .decode("torchrgb") \
    .to_tuple('input.pyd', 'output.pyd') \
    .map_tuple(transform, lambda x: x) \
    .batched(mini_batch_size) \
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None, \
    num_workers=num_workers, \
    persistent_workers=persistent_workers, \
    pin_memory=pin_memory, \
    prefetch_factor=prefetch_factor \
    ).slice(nbatches)

train_loader.length = nbatches
```

← shuffling samples per process

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len) } description of shard content

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)
    } transforming and batching

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len) ← define len(train_dataset)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

batching handled by
WebDataset class

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

} usual DataLoader args

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)  ← drop_last equivalent

train_loader.length = nbatches
```

WebDataset - Implementation

```
import webdataset as wds

def my_splitter(paths):
    paths = list(paths)
    return paths[idr_torch.rank::idr_torch.world_size]

paths = os.environ['DSDIR']+'/imagenet/webdataset/imagenet_train-{000000..000127}.tar'
train_dataset_len = 1281167
train_dataset = wds.WebDataset(paths, nodesplitter=my_splitter, shardshuffle=True)\
    .shuffle(1000)\
    .decode("torchrgb")\
    .to_tuple('input.pyd', 'output.pyd')\
    .map_tuple(transform, lambda x: x)\
    .batched(mini_batch_size)\
    .with_length(train_dataset_len)

nbatches = train_dataset_len // global_batch_size
train_loader = wds.WebLoader(train_dataset, batch_size=None,\
    num_workers=num_workers,\
    persistent_workers=persistent_workers,\
    pin_memory=pin_memory,\
    prefetch_factor=prefetch_factor\
    ).slice(nbatches)

train_loader.length = nbatches
```

← define len(train_loader)

WebDataset - Performance test



I/O loop over the dataset
(calculation-free iterations)

```
start_time = datetime.datetime.now()

for i, (images, labels) in enumerate(loader):
    print(f'{i} / {nb_batches}', end="\r")

end_time = datetime.datetime.now()
delta_time = (end_time - start_time).total_seconds()
```

- Execution on 1 GPU

WebDataset - Performance test

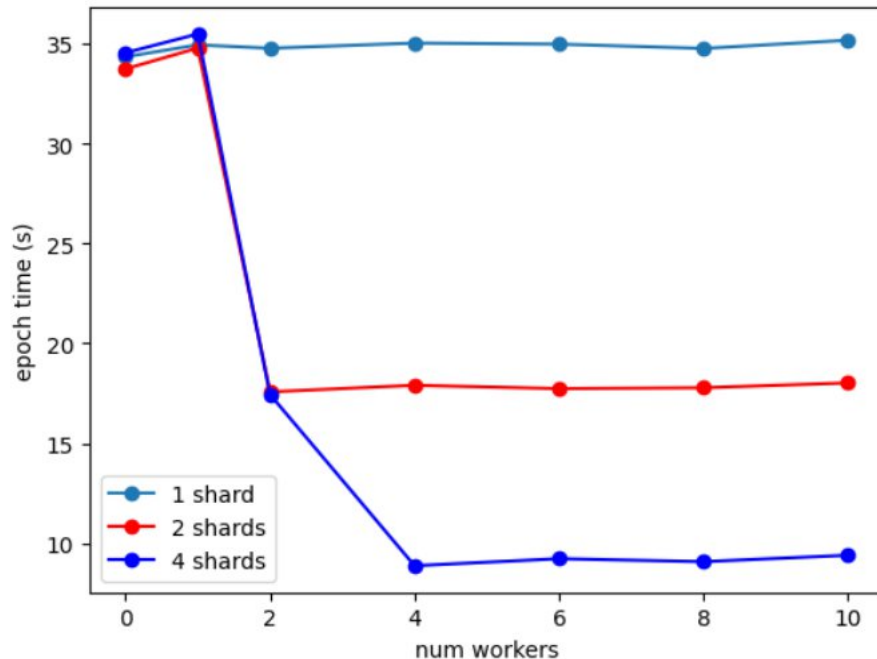


I/O loop over the dataset
(calculation-free iterations)

CIFAR10 ~ 50k images

- Sharding is necessary to benefit from parallel implementation (DataLoader multi-processing).

CIFAR 10 (50k images)
elapsed time - 1 epoch



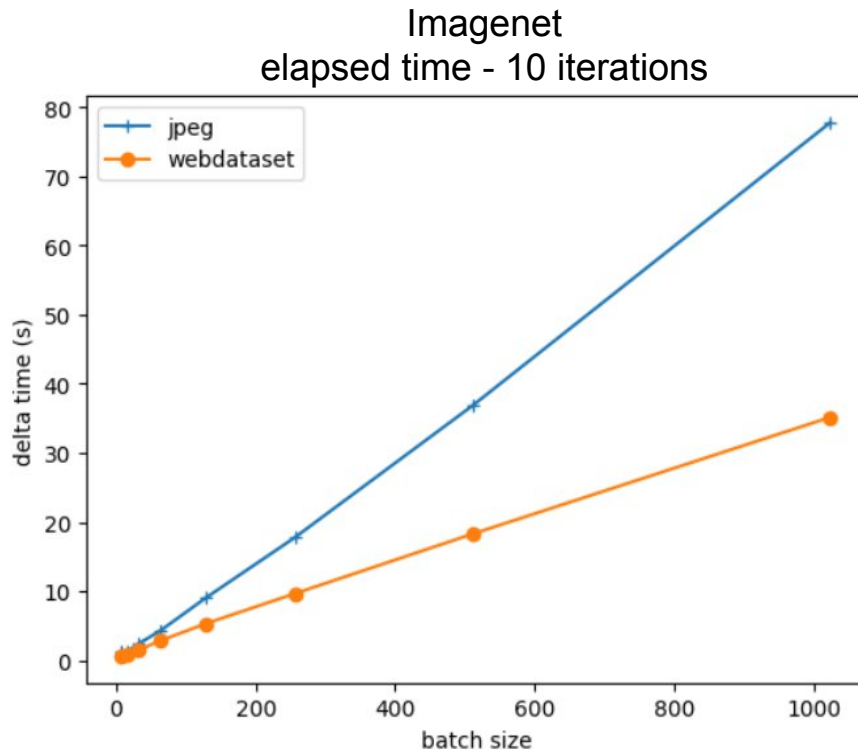
WebDataset - Performance test



I/O loop over the dataset
(calculation-free iterations)

Imagenet ~ 1.3M images
128 shards ~10k images per shard (+labels)
1 shard (images + labels) ~ 6GB

- The more samples are needed per batch, the more efficient is the WebDataset format (fewer I/Os).



———— more I/O generated —————→

WebDataset - Performance test

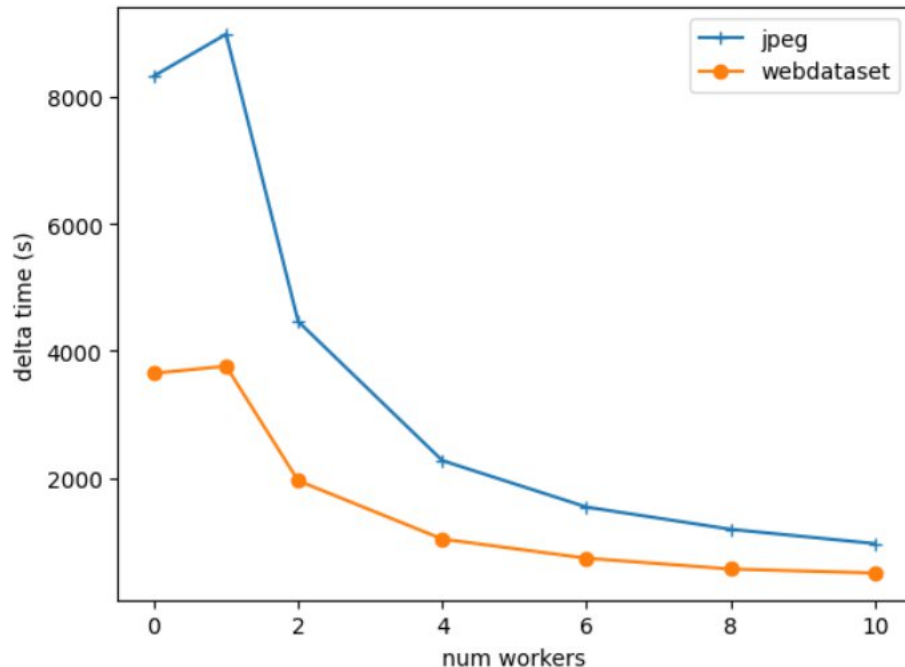


I/O loop over the dataset
(calculation-free iterations)

Imagenet ~ 1.3M images
128 shards ~10k images per shard (+labels)
1 shard (images + labels) ~ 6GB

- The WebDataset format scales up.

Imagenet
elapsed time - 1 epoch

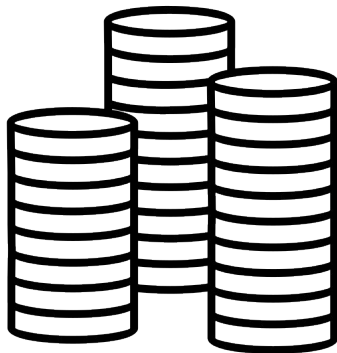


Conclusion on dataset format

- The more inodes your dataset contains, the more IOs you will request during training, and the more difficulty the parallel file system will have
- For large datasets, gather several samples in archives to minimize the number of inodes
- Standard formats already exist like [WebDataset](#) (tar files)

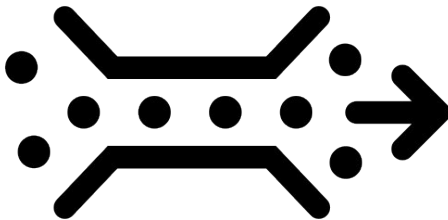
Conclusion

Storage disks



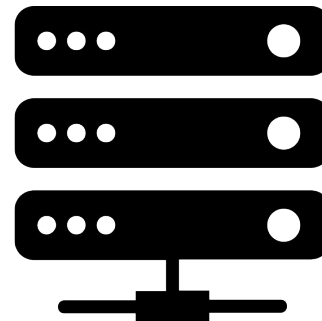
1. I/O performance

Interconnection network Omnipath/InfiniBand



2. Shared bandwidth

CPU workers



3. Decoder performance

- Disk spaces: WORK or SCRATCH ?
- Data format: binary or compressed ?
- Dataset format: WebDataset format ?